CSCI/ECEN 5673: Distributed Systems
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Programming Assignment Three

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Github: https://github.com/anirudhragam/distributed-systems-programming-assignments

Experiments Setup

For this assignment, we used 4 VMs and distributed the buyer/seller server and product/customer DB replicas in the following manner:

VM	Container/ Replica	Protocol : Port
VM1	product-db-0	PostgreSQL: 5432 gRPC: 50051 Raft: 12345
	product-db-1	PostgreSQL: 5433 gRPC: 50054 Raft: 12346
	customer-db-0	PostgreSQL: 5434 gRPC: 50052 ABP UDP: 5100
	customer-db-1	PostgreSQL: 5435 gRPC: 50053 ABP UDP: 5101
	financial-transactions	HTTP: 8000
	seller-server-0	REST: 5000
	buyer-server-0	REST: 6000
VM2	product-db-2	PostgreSQL: 5432 gRPC: 50051 Raft: 12345
	customer-db-2	PostgreSQL: 5433 gRPC: 50052 ABP UDP: 5100
	seller-server-1	REST: 5000
	buyer-server-1	REST: 6000

VM3	product-db-3	PostgreSQL: 5432 gRPC: 50051 Raft: 12345
	customer-db-3	PostgreSQL: 5433 gRPC: 50052 ABP UDP: 5100
	seller-server-2	REST: 5000
	buyer-server-2	REST: 6000
VM4	product-db-4	PostgreSQL: 5432 gRPC: 50051 Raft: 12345
	customer-db-4	PostgreSQL: 5433 gRPC: 50052 ABP UDP: 5100
	seller-server-3	REST: 5000
	buyer-server-3	REST: 6000

On the `test-runner-vm`, we ran the `performance_tests.py` file to run our experiments which we created for PA1.

In the performance tests, we define the `run_buyer_operations` function to run the 1000 buyer operations and the `run_seller_operations` to run 1000 buyer operations. These functions record a list of response times for each operation and return the list of 1000 response times to the caller.

The `run_scenario` function creates one thread for each seller and buyer (1, 10, or 100) in the experiment. Each thread calls its corresponding run operations function to run 1000 operations. The results of the threads are collected as the threads complete. These results are the list of response times for the 1000 operations. The `run_scenario` function creates a list of all seller response times and a list of all buyer response times.

The `compute_metrics` function takes a list of response times as input and computes the average response time and the throughput. Once all the threads have completed, `run_scenario` function calls `computer_metrics` for seller response times and buyer response times to get the average response time and throughput for one run for all sellers and buyers for a scenario.

The `run_experiments` function runs the experiment for 10 iterations. Finally it computes the overall average of the average response times and throughputs for buyers and sellers for the 10 iterations and reports them as the final results of the experiment.

`run_experiment` is called for 1, 10 and 100 buyers and sellers.

Experiment Results

Failure Condition A: All replicas running (baseline)

Scenario 1: Run 1 seller and 1 buyer.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	247.11 ms	4.05 ops/sec	158.60 ms	6.31 ops/sec
Iteration 2	244.31 ms	4.09 ops/sec	167.20 ms	5.98 ops/sec
Iteration 3	230.07 ms	4.35 ops/sec	163.04 ms	6.13 ops/sec
Iteration 4	240.49 ms	4.16 ops/sec	191.05 ms	5.23 ops/sec
Iteration 5	241.94 ms	4.13 ops/sec	179.34 ms	5.58 ops/sec
Iteration 6	249.07 ms	4.01 ops/sec	187.06 ms	5.35 ops/sec
Iteration 7	253.48 ms	3.95 ops/sec	176.27 ms	5.67 ops/sec
Iteration 8	257.91 ms	3.88 ops/sec	171.90 ms	5.82 ops/sec
Iteration 9	256.22 ms	3.90 ops/sec	191.75 ms	5.22 ops/sec
Iteration 10	237.98 ms	4.20 ops/sec	176.18 ms	5.68 ops/sec
Average	245.86 ms	4.07 ops/sec	176.24 ms	5.70 ops/sec

Scenario 2: Run 10 concurrent sellers and 10 concurrent buyers.

	Sellers		Buyers	
	Average Response Time	Throughput (ops/sec)	Average Response Time	Throughput (ops/sec)

	(ms)		(ms)	
Iteration 1	225.56 ms	4.43 ops/sec	107.85 ms	9.27 ops/sec
Iteration 2	336.15 ms	2.97 ops/sec	211.38 ms	4.73 ops/sec
Iteration 3	436.94 ms	2.29 ops/sec	342.16 ms	2.92 ops/sec
Iteration 4	544.40 ms	1.84 ops/sec	450.72 ms	2.22 ops/sec
Iteration 5	600.87 ms	1.66 ops/sec	533.75 ms	1.87 ops/sec
Iteration 6	657.44 ms	1.52 ops/sec	607.22 ms	1.65 ops/sec
Iteration 7	700.43 ms	1.43 ops/sec	662.25 ms	1.51 ops/sec
Iteration 8	768.79 ms	1.30 ops/sec	732.38 ms	1.37 ops/sec
Iteration 9	841.08 ms	1.19 ops/sec	831.67 ms	1.20 ops/sec
Iteration 10	968.94 ms	1.03 ops/sec	966.96 ms	1.03 ops/sec
Average	608.06 ms	1.97 ops/sec	544.63 ms	2.78 ops/sec

Scenario 3: Run 100 concurrent sellers and 100 concurrent buyers.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	178.57 ms	5.60 ops/sec	90.35 ms	11.07 ops/sec
Iteration 2	178.17 ms	5.61 ops/sec	102.55 ms	9.75 ops/sec
Iteration 3	179.14 ms	5.58 ops/sec	182.88 ms	5.47 ops/sec
Iteration 4	180.76 ms	5.53 ops/sec	139.67 ms	7.16 ops/sec
Iteration 5	180.74 ms	5.53 ops/sec	155.90 ms	6.41 ops/sec
Iteration 6	179.55 ms	5.57 ops/sec	161.66 ms	6.19 ops/sec
Iteration 7	180.93 ms	5.53 ops/sec	184.74 ms	5.41 ops/sec
Iteration 8	181.79 ms	5.50 ops/sec	183.20 ms	5.46 ops/sec

Iteration 9	179.59 ms	5.57 ops/sec	197.43 ms	5.07 ops/sec
Iteration 10	180.85 ms	5.53 ops/sec	210.14 ms	4.76 ops/sec
Average	180.01 ms	5.56 ops/sec	160.85 ms	6.67 ops/sec

Failure Condition B: One seller-server and one buyer-server fail

Scenario 1: Run 1 seller and 1 buyer.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	241.96 ms	4.13 ops/sec	176.15 ms	5.68 ops/sec
Iteration 2	247.35 ms	4.04 ops/sec	167.66 ms	5.96 ops/sec
Iteration 3	242.46 ms	4.12 ops/sec	176.32 ms	5.67 ops/sec
Iteration 4	246.63 ms	4.05 ops/sec	187.63 ms	5.33 ops/sec
Iteration 5	253.49 ms	3.94 ops/sec	168.88 ms	5.92 ops/sec
Iteration 6	249.95 ms	4.00 ops/sec	181.75 ms	5.50 ops/sec
Iteration 7	254.87 ms	3.92 ops/sec	183.31 ms	5.46 ops/sec
Iteration 8	253.07 ms	3.95 ops/sec	193.16 ms	5.18 ops/sec
Iteration 9	263.60 ms	3.79 ops/sec	197.37 ms	5.07 ops/sec
Iteration 10	265.02 ms	3.77 ops/sec	214.38 ms	4.66 ops/sec
Average	251.84 ms	3.97 ops/sec	184.66 ms	5.44 ops/sec

Scenario 2: Run 10 concurrent sellers and 10 concurrent buyers.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)

Iteration 1	204.85 ms	4.88 ops/sec	97.19 ms	10.29 ops/sec
Iteration 2	328.68 ms	3.04 ops/sec	205.85 ms	4.86 ops/sec
Iteration 3	446.71 ms	2.24 ops/sec	355.18 ms	2.82 ops/sec
Iteration 4	515.58 ms	1.94 ops/sec	442.33 ms	2.26 ops/sec
Iteration 5	599.84 ms	1.67 ops/sec	521.01 ms	1.92 ops/sec
Iteration 6	639.29 ms	1.56 ops/sec	592.06 ms	1.69 ops/sec
Iteration 7	677.58 ms	1.48 ops/sec	650.69 ms	1.54 ops/sec
Iteration 8	747.02 ms	1.34 ops/sec	728.73 ms	1.37 ops/sec
Iteration 9	842.84 ms	1.19 ops/sec	859.75 ms	1.16 ops/sec
Iteration 10	935.59 ms	1.07 ops/sec	967.85 ms	1.03 ops/sec
Average	593.80 ms	2.04 ops/sec	542.06 ms	2.89 ops/sec

Scenario 3: Run 100 concurrent sellers and 100 concurrent buyers.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	180.18 ms	5.55 ops/sec	76.77 ms	13.03 ops/sec
Iteration 2	178.39 ms	5.61 ops/sec	86.63 ms	11.54 ops/sec
Iteration 3	179.19 ms	5.58 ops/sec	102.31 ms	9.77 ops/sec
Iteration 4	180.74 ms	5.53 ops/sec	130.44 ms	7.67 ops/sec
Iteration 5	179.47 ms	5.57 ops/sec	133.43 ms	7.49 ops/sec
Iteration 6	182.14 ms	5.49 ops/sec	141.22 ms	7.08 ops/sec
Iteration 7	180.77 ms	5.53 ops/sec	151.78 ms	6.59 ops/sec
Iteration 8	182.40 ms	5.48 ops/sec	168.72 ms	5.93 ops/sec

Iteration 9	180.30 ms	5.55 ops/sec	173.67 ms	5.76 ops/sec
Iteration 10	180.54 ms	5.54 ops/sec	184.46 ms	5.42 ops/sec
Average	180.41 ms	5.54 ops/sec	134.94 ms	8.03 ops/sec

Failure Condition C: One non-leader product-db replica fails
Scenario 1: Run 1 seller and 1 buyer.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	590.78 ms	1.69 ops/sec	18.81 ms	53.15 ops/sec
Iteration 2	657.88 ms	1.52 ops/sec	92.83 ms	10.77 ops/sec
Iteration 3	598.23 ms	1.67 ops/sec	101.50 ms	9.85 ops/sec
Iteration 4	560.23 ms	1.78 ops/sec	105.17 ms	9.51 ops/sec
Iteration 5	676.61 ms	1.48 ops/sec	158.41 ms	6.31 ops/sec
Iteration 6	572.84 ms	1.75 ops/sec	138.79 ms	7.21 ops/sec
Iteration 7	490.68 ms	2.04 ops/sec	105.89 ms	9.44 ops/sec
Iteration 8	572.75 ms	1.75 ops/sec	180.17 ms	5.55 ops/sec
Iteration 9	656.14 ms	1.52 ops/sec	131.77 ms	7.59 ops/sec
Iteration 10	595.68 ms	1.68 ops/sec	58.47 ms	17.10 ops/sec
Average	597.18 ms	1.69 ops/sec	109.18 ms	13.65 ops/sec

Scenario 2: Run 10 concurrent sellers and 10 concurrent buyers.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)

Iteration 1	560.02 ms	1.79 ops/sec	147.65 ms	6.77 ops/sec
Iteration 2	583.04 ms	1.72 ops/sec	137.44 ms	7.28 ops/sec
Iteration 3	541.43 ms	1.85 ops/sec	120.69 ms	8.29 ops/sec
Iteration 4	600.64 ms	1.66 ops/sec	130.66 ms	7.65 ops/sec
Iteration 5	580.71 ms	1.72 ops/sec	134.11 ms	7.46 ops/sec
Iteration 6	575.79 ms	1.74 ops/sec	147.18 ms	6.79 ops/sec
Iteration 7	591.02 ms	1.69 ops/sec	149.59 ms	6.69 ops/sec
Iteration 8	576.63 ms	1.73 ops/sec	145.37 ms	6.88 ops/sec
Iteration 9	594.11 ms	1.68 ops/sec	147.51 ms	6.78 ops/sec
Iteration 10	526.11 ms	1.90 ops/sec	152.11 ms	6.57 ops/sec
Average	572.95 ms	1.75 ops/sec	141.23 ms	7.12 ops/sec

Scenario 3: Run 100 concurrent sellers and 100 concurrent buyers.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	543.90 ms	1.84 ops/sec	32.41 ms	30.85 ops/sec
Iteration 2	554.36 ms	1.80 ops/sec	25.97 ms	38.51 ops/sec
Iteration 3	549.79 ms	1.82 ops/sec	28.36 ms	35.26 ops/sec
Iteration 4	535.90 ms	1.87 ops/sec	44.02 ms	22.72 ops/sec
Iteration 5	587.60 ms	1.70 ops/sec	35.67 ms	28.04 ops/sec
Iteration 6	569.53 ms	1.76 ops/sec	33.99 ms	29.42 ops/sec
Iteration 7	538.39 ms	1.86 ops/sec	32.74 ms	30.55 ops/sec
Iteration 8	592.50 ms	1.69 ops/sec	28.53 ms	35.05 ops/sec

Iteration 9	585.30 ms	1.71 ops/sec	31.10 ms	32.15 ops/sec
Iteration 10	578.09 ms	1.73 ops/sec	30.83 ms	32.44 ops/sec
Average	563.54 ms	1.78 ops/sec	32.36 ms	31.50 ops/sec

Failure Condition D: Leader product-db replica fails
Scenario 1: Run 1 seller and 1 buyer.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	511.65 ms	1.95 ops/sec	103.75 ms	9.64 ops/sec
Iteration 2	481.14 ms	2.08 ops/sec	56.17 ms	17.80 ops/sec
Iteration 3	430.53 ms	2.32 ops/sec	70.85 ms	14.11 ops/sec
Iteration 4	600.96 ms	1.66 ops/sec	146.26 ms	6.84 ops/sec
Iteration 5	526.96 ms	1.90 ops/sec	138.13 ms	7.24 ops/sec
Iteration 6	658.06 ms	1.52 ops/sec	161.45 ms	6.19 ops/sec
Iteration 7	559.83 ms	1.79 ops/sec	167.06 ms	5.99 ops/sec
Iteration 8	571.47 ms	1.75 ops/sec	170.80 ms	5.85 ops/sec
Iteration 9	635.04 ms	1.57 ops/sec	157.07 ms	6.37 ops/sec
Iteration 10	656.23 ms	1.52 ops/sec	147.94 ms	6.76 ops/sec
Average	563.19 ms	1.81 ops/sec	131.95 ms	8.68 ops/sec

Scenario 2: Run 10 concurrent sellers and 10 concurrent buyers.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)

Iteration 1	597.16 ms	1.67 ops/sec	135.04 ms	7.41 ops/sec
Iteration 2	678.45 ms	1.47 ops/sec	239.79 ms	4.17 ops/sec
Iteration 3	812.57 ms	1.23 ops/sec	402.37 ms	2.49 ops/sec
Iteration 4	893.56 ms	1.12 ops/sec	554.41 ms	1.80 ops/sec
Iteration 5	976.83 ms	1.02 ops/sec	674.02 ms	1.48 ops/sec
Iteration 6	1049.47 ms	0.95 ops/sec	783.96 ms	1.28 ops/sec
Iteration 7	1167.84 ms	0.86 ops/sec	891.93 ms	1.12 ops/sec
Iteration 8	1294.33 ms	0.77 ops/sec	1015.52 ms,	0.98 ops/sec
Iteration 9	1394.60 ms	0.72 ops/sec	1182.86 ms	0.85 ops/sec
Iteration 10	1511.83 ms	0.66 ops/sec	1315.26 ms	0.76 ops/sec
Average	1037.66 ms	1.05 ops/sec	719.52 ms	2.23 ops/sec

Scenario 3: Run 100 concurrent sellers and 100 concurrent buyers.

	Sellers		Buyers	
	Average Response Time (ms)	Throughput (ops/sec)	Average Response Time (ms)	Throughput (ops/sec)
Iteration 1	561.63 ms	1.78 ops/sec	150.96 ms	6.62 ops/sec
Iteration 2	560.39 ms	1.78 ops/sec	159.84 ms	6.26 ops/sec
Iteration 3	551.47 ms	1.81 ops/sec	179.72 ms	5.56 ops/sec
Iteration 4	570.04 ms	1.75 ops/sec	217.88 ms	4.59 ops/sec
Iteration 5	562.09 ms	1.78 ops/sec	257.50 ms	3.88 ops/sec
Iteration 6	563.02 ms	1.78 ops/sec	301.36 ms	3.32 ops/sec
Iteration 7	562.04 ms	1.78 ops/sec	379.50 ms	2.64 ops/sec
Iteration 8	592.78 ms	1.69 ops/sec	479.64 ms	2.08 ops/sec

Iteration 9	712.27 ms	1.40 ops/sec	757.75 ms	1.32 ops/sec
Iteration 10	1358.61 ms	0.74 ops/sec	502.23 ms	1.99 ops/sec
Average	659.43 ms	1.63 ops/sec	338.64 ms	3.83 ops/sec

Results Explanation

Comparisons between the three scenarios

- Across all three scenarios, the results for Failure Condition A (no failures) and B (stop one seller server and one buyer server) were approximately the same. This suggests that the cost of the clients hitting the stopped server 0s and then failing over to the other replicas is minimal.
- The difference in Average Response time between failure condition C (non-leader replica failing) and failure condition D (leader replica failing) is small. This is expected as Raft leader election is relatively fast.
- In general, as the number of concurrent clients increased, the average response time increased and throughput decreased which is expected because the load on the servers and databases increases.

Most of the executions were run on GCE instances in us-west-1a. However due to VM unavailability, some executions were run in us-west-1c, which introduced slight inconsistencies in the data.

Comparison between PA2 and PA3 results

PA3 introduces rotating-sequencer atomic broadcast (ABP) for the customer database and Raft for the product database, both of which require quorum of replicas for consensus. This coordination overhead raises the average response times compared to PA2 where each database was a single replica that could commit writes immediately and did not require quorum.